# Shubhadeep Mandal

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## Education

**International Institute of Information Technology, Hyderabad** — July 2025 – Present
*Telangana, India*
*Master of Technology in Computer Science and Engineering* — *CGPA: 9.12*

**Indian Institute of Information Technology, Ranchi** — Aug 2020 – Jun 2024
*Ranchi, India*
*Bachelor of Technology (Hons.) in Computer Science and Engineering* — *CGPA: 9.05*

## Projects

**Negotiate, Then Solve: Full-Stack LLM Scheduling Platform** | *Python, FastAPI, React, SQLite, RAG*

- Built a React 19 + TypeScript front end over a FastAPI REST API with server-enforced role-based permissions, an approval queue, audit traces and version rollback, persisted in SQLite and backed by **215 pytest tests**.
- Built an agentic workflow that turns free-text requests from email, chat and a web portal into typed constraints, resolving conflicts via CP-SAT analysis and LLM negotiation through typed tool calls.
- Built hybrid BM25 + dense-embedding retrieval over OCR-ingested PDFs and web pages, raising policy Recall@5 to **93.4%** (vs 87.7% BM25); guardrails ensured **no attack succeeded** across 50 prompt-injection and spoofing tests.
- Added Gemini API key pooling with failover and rate limiting, and a solver cache cutting demo replay from **2.5 min to 25 s**.

**Distributed Data Pipelines: MapReduce & gRPC Streaming** | *Python, gRPC, Hadoop, C++*

- Built a real-time gRPC streaming pipeline (coordinator + worker services over bidirectional streams) sustaining **96–123K records/s**, answering live analytics queries in 2.1–4.5 ms with backpressure and **zero record loss**.
- Collapsed a 19-metric weather analytics batch job into a single Hadoop MapReduce stage where no raw record crosses the shuffle, byte-identical to sequential and MPI baselines.
- Validated every pipeline with **66 automated checks** that start real servers and compare outputs byte for byte against reference implementations.

**Warehouse Traffic Control: Distributed Robot Coordination** | *JavaScript, Three.js, Web Workers*

- Designed a lease-and-fencing protocol coordinating 200 robots across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Implemented deadlock detection and crash recovery that rebuilds a manager's state in ∼4.3 s, plus gossip congestion control that raised tasks completed by **98%** at 150 robots.
- Versus one central coordinator: **7.6x lower peak per-node load** (125 vs 950 msgs/s), and a region-manager crash cuts off **only 11 robots instead of all 120**; deployed as a live 3D simulator via GitHub Actions.

**IndicLLM: Transformer Pretraining & Reasoning SFT** | *PyTorch, ONNX, Docker*

- Built Hindi and Assamese LMs end to end: curated 1.25B tokens (20%+ self-crawled and OCR'd), trained native 16K BPE tokenizers, and pretrained four 25M-param Transformers from PyTorch primitives on 500M tokens each.
- Fine-tuned on solver-generated reasoning, reaching **0.96** test accuracy and **0.85** on 10-hop chains (3× the training depth) vs a 0.48 baseline, with a leakage check ruling out answer lookup.
- Exported all 10 models to ONNX with a KV cache and a JavaScript tokenizer so they run fully in the browser, deployed as a live demo on Railway via Docker and FastAPI.

## Technical Skills

**Languages**: Python, C/C++, JavaScript, TypeScript, SQL, Rust, Bash; OOP, Data Structures & Algorithms.

**Backend & Data**: FastAPI, REST APIs, gRPC, Protocol Buffers, SQLite, PostgreSQL, Hadoop MapReduce, YARN.

**AI & ML**: LLM Agents, RAG, Prompt Engineering, LoRA, Pretraining & SFT, PyTorch, ONNX, scikit-learn.

**Web & DevOps**: React, Next.js, Three.js, Git, GitHub Actions CI/CD, Docker, Railway, Vercel, pytest, Linux.

## Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

## Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.